Introduction to 3D Printing Center

3D Printing Center - Village School

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6-Lesson Upper Grade Center Curriculum (Grades 3-5)

Course Overview

Students will learn basic 3D design and printing through hands-on projects using Tinkercad. This 6-lesson rotation (45 minutes each) begins with understanding what 3D printing is, then progresses to hands-on design and creation of simple, functional objects.

Learning Objectives

Students will:

- Understand what 3D printing is and how it works
- Navigate Tinkercad to create basic 3D designs
- Design objects considering size and printing limitations
- Create at least two original 3D designs
- Identify real-world applications of 3D printing

Materials Needed

- 3D printer (any modern FDM printer)
- PLA filament
- Chromebooks with internet access
- Sample printed objects for demonstration
- Cross-section examples of printed objects

Reworks for second rotation

Lessons 3,4,5,6 will be reworked to have more specific objectives.

Lesson 2 is the first Tinkercad lesson, and it will continue to use the keychain project. I am tempted to switch it over to the more specific approach to making the keychain here: https://www.tinkercad.com/projects/Name-That-Keychain-1

Lesson 3: Create a 20cm long ruler, accurate to the nearest millimeter. This project will reinforce their connection of dimensions between Tinkercad world and real world. It will have them using the duplicate tool and probably the ruler tool in Tinkercad. There's some room for embelishments but the main goal is common for all the students.

Lesson 4: Create something with joints – introduce the articulating joints in Tinkercad

Lesson 5: 3D printed zipper pulls https://www.tinkercad.com/projects/3D-Printed-Zipper-Pulls

Lesson 6: The challenge with this lesson is that there is no good opportunity for them to pick up things they design after the lesson. However doing a show and tell reflection kind of lesson didn't work out as we found last time. An idea I'm contemplating is to have them assemble a device out of 3D printed parts that will be ready on that day. I'm open to ideas on objects that can be assembled, but i'm tempted to use one of the objects that the YouTuber Engineezy created.

Examples:

- 2DOF Robot Head https://engineezy.com/collections/free-downloads/products/robbob-2-dof-robot-head
- Animation Fidget https://engineezy.com/collections/free-downloads/products/the-animation-fidget-stl
- Stomp Rocket (Would use a soft filament and remove the pointy tip) https://engineezy.com/collections/free-downloads/products/the-epic-stomp-rocket-stl
- Ellipse Drawing Mechanism https://engineezy.com/collections/free-downloads/products/the-ellipse-mechanism

Lesson 1: What is 3D Printing? (45 minutes)

Establish class expectations (5 minutes)

For example but not limited to:

- Model Respect
  - respect centermates' work
  - respect the printers
    - don't get hurt by them
    - don't hurt them
  - respect the mere 45 minutes per center lesson
- Make Good Choices
- Solve Problems
  - If you know how, help your peers

Introduction and Hook (5 minutes)

Hook: Show students amazing 3D printed objects - toys, tools, food!

Key Question: "What is 3D printing and how does it work?"

Core Concepts with Visual Demonstrations (10 minutes)

How 3D Printing Works:

- 3D printing builds objects layer by layer, like stacking pancakes
- We design on computers, then the printer makes it real
- Plastic gets heated and squeezed out like frosting from a tube
- Show infographic from Placerville EDC Library

Visual Demonstrations:

- Use simple infographics showing the layer-by-layer process
- Pass around printed objects at different stages (if available)
- Show cross-section of a printed object to reveal layers

Real-World Examples and Applications (10 minutes)

Examples for Kids:

- Custom phone cases and game pieces
- Replacement parts for broken toys
- Art projects and sculptures
- Helpful tools
- Aerospace and car parts
- Food printing (chocolate!)

Interactive Activity: Students examine printed objects and guess what they're used for

3D Printer Live Demonstration (10 minutes)

- Show the actual 3D printer (if available)
- Explain the basic parts: hot end, build plate, filament
- Safety rules: Never touch hot parts, don't reach in while printing (pinch hazards)

Cleanup and Preview (5 minutes)

- Students help organize demonstration materials
- Preview next lesson: "Now that we know what 3D printing is, we'll learn to design our own objects!"
- Quick reflection: "What was the most surprising thing you learned about 3D printing?"

Lesson 2: Introduction to Tinkercad + Design Your First Project (45 minutes)

Introduction to Tinkercad (15 minutes)

1. **Login to Tinkercad** (5 minutes)
   - Use school accounts
   - Navigate to tinkercad.com
2. **"Learning the Moves" Tutorial** (10 minutes)
   - Built-in Tinkercad tutorial (abbreviated version)
   - Focus on basic navigation: rotate, zoom, pan
   - Basic shape placement and manipulation

First Design Project: Simple Keychain (20 minutes)

Project Requirements:

- Must have a hole for keyring
- No bigger than 50mm in any direction
- At least 3mm thick
- Use at least 2 shapes

Step-by-Step Guidance:

1. Start with a basic shape (cube, cylinder, etc.)
2. Add text with their name or initials
3. Create a hole for the keyring
4. Combine shapes together

Teacher Support:

- Circulate constantly - students need lots of help with first design
- Have printed keychain examples available
- Focus on successful completion rather than complexity

Save, Share, and Cleanup (10 minutes)

- Students save their work (5 minutes)
- Quick gallery walk to see designs (3 minutes)
- Cleanup workstations and preview next lesson (2 minutes)

Lesson 3: How Printing Works + Design a Container (45 minutes)

Printing Process Deep Dive (10 minutes)

- Review keychains that have been printed from last lesson
- Explain the printing process in detail (in a show and tell way with Polar3D/Bambu Studio):
  - Computer slices design into thin layers (show slicing software if possible)
  - Printer draws each layer with melted plastic
  - Layers stick together as they cool
- Show printer in action if available

Safety Rules Review:

- Never touch the hot parts
- Don't reach into the printer while it's working
- Tell a teacher if something looks wrong

New Project: Simple Container (25 minutes)

Design Challenge: Create a small container for desk supplies

Requirements:

- Big enough to hold paperclips or erasers
- Has walls and a bottom
- Learn to use the "Hole" tool to hollow out the inside

Step-by-Step Guidance:

1. Start with a cube or cylinder
2. Add a smaller shape inside
3. Make the inner shape a "Hole"
4. Group them together

Troubleshooting and Cleanup (10 minutes)

- Help students who are struggling
- Save work and clean up workstations

Lesson 4: Problem-Solving Design Challenge (45 minutes)

Problem Introduction (10 minutes)

The Challenge: "Design something to solve a problem at school or home"

Examples to spark ideas:

- Pencil holder that doesn't tip over
- Pencil topper
- Pencilcase
- Hook for backpack
- Bookmark that stays in place
- Cord organizer for headphones
- Organizer for small toys or supplies

Design Process (30 minutes)

1. **Think** (5 minutes): What problem will you solve?
2. **Sketch** (5 minutes): Quick sketch on paper
3. **Build** (20 minutes): Create in Tinkercad

Teacher Role:

- Help students identify realistic problems
- Guide them toward printable solutions
- Encourage creativity while keeping designs simple

Sharing and Cleanup (5 minutes)

- Students briefly explain their problem and solution (3 minutes)
- Clean up workstations (2 minutes)

Lesson 5: Improve Your Design + Print Prep (45 minutes)

Review Printed Projects (10 minutes)

- Look at keychains and containers that have been printed
- Discuss what worked well and what could be better
- Introduce "iteration" - making designs better through improvement

Design Improvement Time (25 minutes)

Students can:

- Fix problems with their existing designs
- Make their problem-solving project better
- Start a completely new project if they want

Focus on:

- Making walls thick enough (at least 2mm)
- Checking that parts fit together
- Adding personal touches and creativity

File Preparation and Cleanup (10 minutes)

- Students export their best design as STL file (5 minutes)
- Teacher demonstrates the export process
- Save files for printing and clean up (5 minutes)

Lesson 6: Reflection and Future Thinking (45 minutes)

Show and Tell (20 minutes)

- Students present their printed objects
- Simple format: "This is my _____, it solves the problem of _____, and I'm proud of _____"
- Celebrate everyone's work!

Reflection Activity (15 minutes)

Discussion Questions:

- What was the hardest part about 3D design?
- What surprised you about 3D printing?
- If you had a 3D printer at home, what would you make?
- How could 3D printing help people?

Quick Write: Students write 2-3 sentences about their favorite project

Looking Forward and Cleanup (10 minutes)

- Show advanced 3D printed objects (gears, moving parts, etc.) (5 minutes)
- Explain that 3D printing is used in many jobs (3 minutes)
- Final cleanup and encourage continued exploration (2 minutes)

Assessment and Differentiation

For 3rd Graders:

- Focus on basic shape manipulation
- Provide more step-by-step guidance
- Celebrate any successful design completion

For 4th Graders:

- Encourage combining multiple shapes
- Introduce basic problem-solving concepts
- Support independent exploration

For 5th Graders:

- Challenge with more complex designs
- Discuss real-world applications
- Encourage peer mentoring of younger students

Quick Assessment Checkpoints:

- Can student navigate Tinkercad independently?
- Does student understand basic 3D printing concept?
- Has student completed at least one printable design?
- Can student explain what they created?

Teacher Preparation Notes

Before Each Lesson:

- **Lesson 1:** Prepare infographics, demonstration objects, and cross-sections
- **Lessons 2-6:** Test all computers/tablets for Tinkercad access
- Have backup activities for students who finish early
- Queue up printing jobs from previous lessons
- Prepare printed examples to show
- **Important:** Plan for 5-minute cleanup time at end of each lesson

Printing Management:

- Print student work between lessons when possible
- Keep designs simple to ensure printing success
- Have a few backup printed items in case of printer failures
- Consider printing 2-3 designs per session to manage time

Common Challenges:

- **Students going too fast:** Encourage them to add details or help peers
- **Students struggling:** Pair with successful peers or provide extra guidance
- **Technical issues:** Have simple backup activities ready
- **Mixed grade levels:** Use peer mentoring and differentiated expectations
- **Time management:** Shorter lessons require tighter pacing and clear transitions

Lesson 1 Specific Preparation:

- Gather diverse 3D printed objects from different industries
- Prepare cross-section examples (cut printed objects in half if possible)
- Have printer ready for live demonstration
- Plan interactive activities for examining objects

Extension Ideas:

- Create a classroom gallery of printed objects
- Connect to other subjects (math shapes, science tools, art projects)
- Invite parents to see student work
- Consider a school-wide 3D printing showcase
- Document the “what is 3D printing” lesson with photos for future reference